

cumulative time of observations. Therefore, the last partial mortality rate resulted from two years of observations (no observations were taken within the interval of 1 to 3 years); the expected PTM in the time interval of 1 - 2 or 2 - 3 would be lower than observed. In addition, coral fragments that died three years after the hurricane event were also those that showed a significant loss of tissue ($p < 0.05$, ANOVA) during the first year ($68.35 \pm 25.63\%$, $n=113$) compared with those that were still alive after the three years ($56.72 \pm 24.70\%$ of tissue lost during the first year, $n=95$). This reduction of live tissue over the fragment had negative effects on the size frequency distribution of the generated fragments. For example, in 1998 the peak of fragment distribution was within the interval of 200 to 400 cm², but one year later, most surviving colonies suffered significant and progressive partial tissue mortality causing the displacement of the fragment size distribution to the smaller size classes (Figure 3.4).

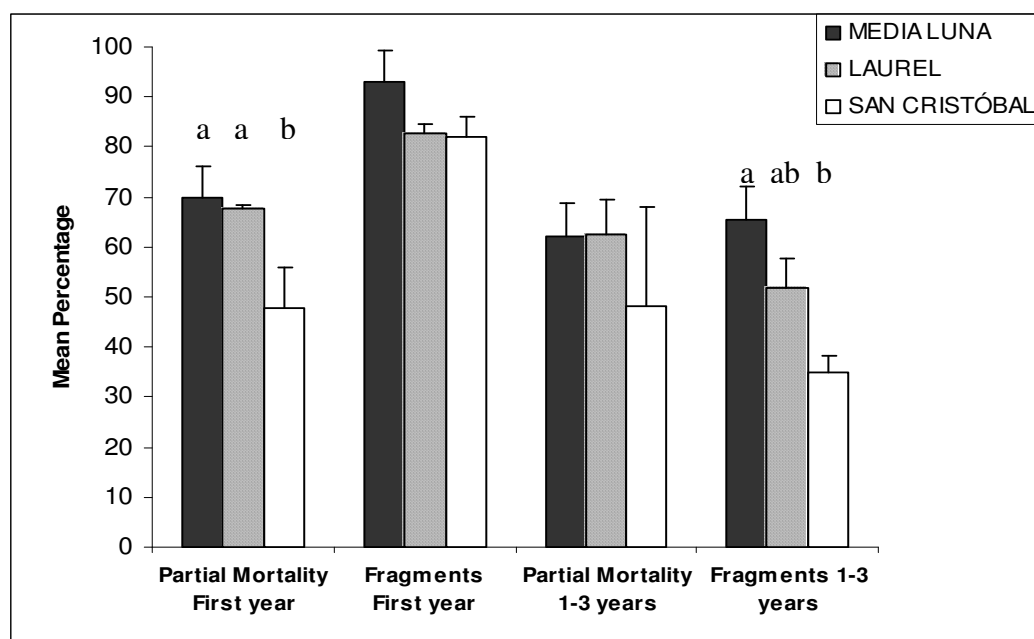


Figure 3.3.- Distribution of mean percentage partial tissue mortality per fragment, and the mean percentage of fragments observed with partial tissue mortality at two time intervals within each reef. Different letters represented significant difference between means ($p < 0.05$, Tukey-Kramer Test). A line above each bar represents one standard deviation.

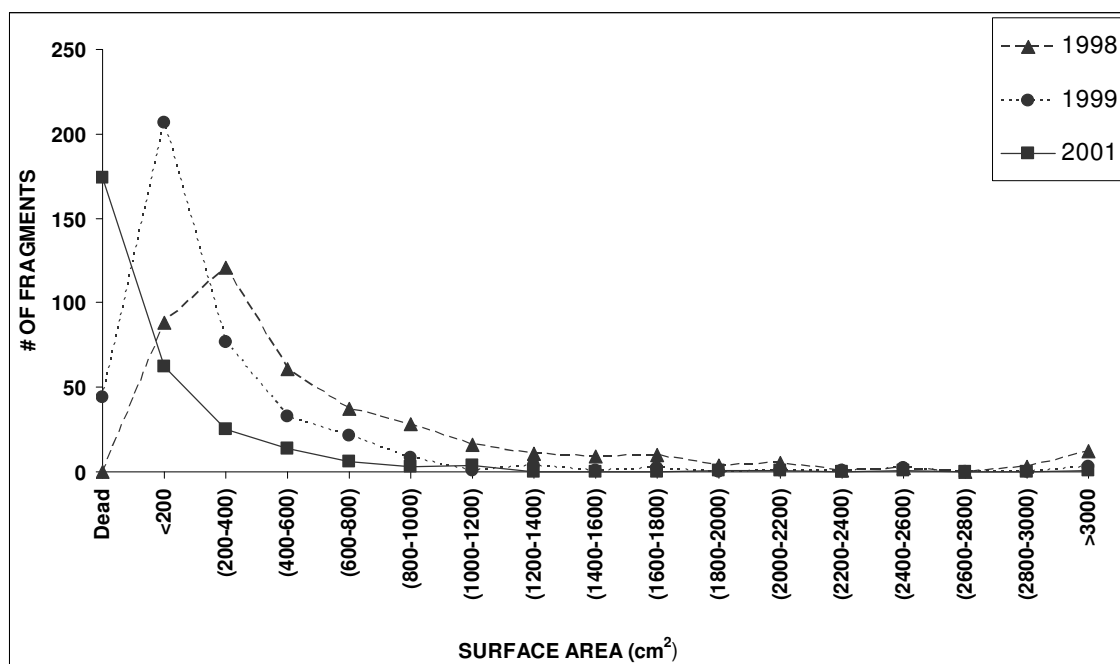


Figure 3.4. - Size frequency distributions of 401 fragments observed at three-time intervals.

3.3.3. - Tissue Growth vs. Tissue Reduction

Colony growth, defined as the increase in live coral surface area, was observed after a year in San Cristóbal Reef and after 3 years in all three reefs (Table 3.2). Within the first year, a small proportion of fragments (13 %) showed an increase of more than 128 % of their live surface area with a total amount of 1,112 cm² of new coral tissue gained by growth. However, this increment was overwhelmed by the large proportion of fragments (81.3 %) that showed a mean decline of more than 46.5 % of their live surface area due to partial tissue mortality. The total amount of lost tissue (16,854 cm²) was 14 times larger when compared to the amount gained by growth. In addition, the mean surface area of lost tissue by fragments was significantly higher ($p < 0.05$, t-test) compared with the mean surface area gained by coral growth. During the time interval of 1 to 3 years, the mean surface area lost by partial tissue mortality was also significantly higher when compared with the mean surface area gained by colony growth